QC – CHECKLIST

Rev1: 2025-04-18

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SOP Number **001**

SOP Title **QC Checklist**

	NAME	**TITLE**	**SIGNATURE**	**DATE**
Author	Manjil Puri	Supervising Engineer		04/18/2025
Reviewer	Jocelyn Ignacia	Electrical Engineer I		07/07/2025
Authoriser				

Effective Date:	
Review Date:	**07/07/2025**

TITLE BLOCK

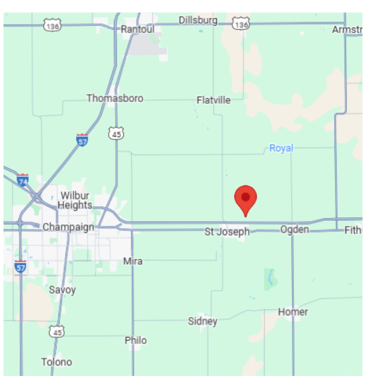
1. SOV, Date, Designer, Engineer, Revision Number
2. EPC Logo, Owner Logo
3. Castillo Project ID
4. Correct EOR Seal
5. Project Name
6. Project Address
7. Site Coordinates
8. DER Number (If Ameren)

COVER SHEET

1. Project Name, Address, Site Coordinates

2. Building Codes – State/County/City Adopted, Castillo Preference – Information Not Available

3. Owner:
   3.1. Name
   3.2. Address
   3.3. Telephone Number
4. EPC:
   4.1. Name
   4.2. Address
   4.3. Telephone Number
5. Castillo:
   5.1. EOR Name, License Number, State
   5.2. Checker and Designer
6. County Map:
   6.1. State Map
   6.2. County Name not blocked by arrow
7. Array View:
   7.1. Image, Modules, Fence, Road
   7.2. North Arrow
8. Vicinity Map
   8.1. Nearest City and Roads to Site



9. Drawing Index
   9.1. Check for consistency

9.2. Check each page for plotting issue and missed page names/Number

9.3. Check Spelling

SYSTEM INFORMATION TABLE

1. PV Module Make and Model
2. Module STC Rating
3. String Size
4. String Quantity
5. Total DC Size
6. Inverter Make and Model
7. Inverter kVA and kW Rating
8. Derated kVA and kW Rating (If applicable)
9. Inverter Quantity
10. Total AC Size
11. Racking Make, Model and Size
12. DC/AC Ratio
13. Pitch
14. Interrow Spacing
15. GCR
16. Tilt / Tracker Rotation Angle
17. Azimuth

GENERAL NOTES

1. Check for Latest Version

SITE PLAN

1. Split Pages per your discretion
2. North Arrow
3. Text size is consistent
4. Show MV Run and CAB Line
5. Properly faded background IMAGE
6. Notes:
   - 6.1. DCB/Inverter Topology
   - 6.2. Weather Sensor and sheet reference
   - 6.3. Sheet Reference for Pole Lineup, Detailed Site Plan etc.
   - 6.4. Fence Grounding note
7. Section View of Racking
   - 7.1. Fixed vs tracker
   - 7.2. Pitch
   - 7.3. Interrow Spacing
   - 7.4. Racking Height
8. Scale
9. Dimensions
   - 9.1. Dimension from property Line
   - 9.2. Dimension from all four corners

9.3. Dimension of setback

9.4. Width and Turn around Radius of Road

9.5. Fence Setback

10. Topo Lines with elevation

11. Callouts:

    11.1. Identify Site Access

    11.2. Fence, Pad Location, Gate Size, Road, Access Pathway Width

    11.3. Racking Size and Qty

    11.4. Property Line, Lease Line, Parcel ID

    11.5. Setbacks: Wetland, Gas Line, Wells etc.

    11.6. Inverter or DCB Typical Note

    11.7. Civil: Vegetative Buffer, Landscape Buffer, Concrete Washout

12. Proper greying for Visibility

Pole Line UP

1. North Arrow and Scale

2. Is the Pole line up ASSUMED? If so, add a note

3. **Pole Name and Number Consistent with SLD, 3LD and Pole Elevation**

4. Correct Legend Table

5. POI:

    5.1. Utility Name **(NGRID?)**

    5.2. Voltage, Feeder Grounding

    5.3. Pole Number, Feeder Number

    5.4. Fault Current: Ultimate vs Exact

    5.5. Labeled as (POINT OF INTERCONNECTION)

6. Compare Poles with Impact Study or Client CAD

7. Dimension:

    7.1. Distance Between Poles

    7.2. Distance from Access Road, Fence, Gate etc.

8. Callouts:

    8.1. UNDERGROUND MV RUN FROM EQUIPMENT AREA

    8.2. OVERHEAD LINES: NEW VS EXISITING

9. DELINIATION LINE FOR CLIENT VS UTILITY

10. Grey Out Adjacent Interconnection

ENGINEERED EQUIPMENT LIST

1. Review submittal status
2. Use Joe's Table for Equipment Naming Format
3. For all equipment check:
    3.1. Equipment Name
    3.2. Quantity
    3.3. Manufacturer
    3.4. Model
    3.5. Rating

4. **THIS INFO SHOULD BE CONSISTENT WITH SLD, 3LD AND THE OVERALL DESIGN**

AC SINGLE LINE DIAGRAM

1. CONSISTENT NAMES AND RATINGS ACROSS SITE PLAN, POLE LINE UP, SLD, 3LD AND POLE ELEVATION.

2. Notes:
    2.1. Feeder Grounding Note
    2.2. Any assumptions that you made
    2.3. Check page references for notes

3. Stringing:
    3.1. Is stringing, Ok? V, I, P, MPPT, No of Inputs, Temperature
    3.2. Strings Distribution in DCB vs Inverter
    3.3. DCB and String Fuse

4. Relevant Equipment Rating and Quantity
    4.1. Did you use Joe's Table for Equipment Naming Format?
    4.2. Consistent Names in SLD/3LD/Site Plan
    4.3. Exact Model and Make
    4.4. Current
    4.5. LV Voltage Notation: (480/277 V, 208/120 V, 120/240 V)
    4.6. Phase
    4.7. SCCR
    4.8. NEMA Rating
    4.9. I/O Cable Size, Sets, Material
    4.10. Max AND Min OCPD Size
    4.11. AC/DC Disconnecting Means
    4.12. No Inputs: Inverter, Switchboard, Aux Panel
    4.13. Breakers: **Phase**, Count, Dial Settings, Cable Limits (Size, Sets, Material)

5. Six Disconnect Rule for Switchboard?

6. Relevant Transformer Rating
    6.1. XFMR kVA >= Total kVA of Inverter
    6.2. Primary and Secondary Winding Configuration
    6.3. Primary and Secondary Voltage, BIL
    6.4. Cooling Type
    6.5. Z%, X/R
    6.6. Transformer Internal Fuse and Switch
    6.7. Does grounding XFMR match with CESIR?

7. MV Equipment
    7.1. Did you use Joe's Table for Equipment Rating?
    7.2. HD Arrestor for Riser
    7.3. Surge Arrestor Rating
    7.4. Solid Blade vs Fuse for Riser Pole
    7.5. Meter Power and Communication
    7.6. Meter CT and VT Ratio
    7.7. Meter Accuracy Class
    7.8. How is the Recloser Powered and Communication Line?
    7.9. Aux Transformer rating for recloser
    7.10. Recloser CT and VT ratio

8. Poles
    8.1. Use POI Callout Details from Site Plan
    8.2. Consistent Names
    8.3. Consistent Ratings
    8.4. Quantity consistent with Site Plan
    8.5. Delineation Line
    8.6. Label GOAB as Main Service Disconnect

9. Cable and Conduit Sizing
    9.1. Is MV Cable Direct Buried?
    9.2. Check cable Insulation
    9.3. Cable Size – Below vs Above 800 A
    9.4. Terminal Rating of Circuit – DC, LV, MV
    9.5. Is cable sets below 10?

DC

(2) #10 AWG PV WIRE CU
90°C RATED WIRE
IN MIXED RACKING AND 1" CONDUIT

(2) 500 KCMIL PV WIRE AL
(1) #3 AWG PV WIRE - CU EGC
@90°C RATED WIRE
IN 3" CONDUIT

LV

(3) 4/0 AWG XHHW-2 AL
(1) #4 AWG XHHW-2 AL EGC
@90°C RATED WIRE
IN 2" CONDUIT

(10) SETS OF (4) 600 kCMIL THWN-2 CU,
(1) 500 kCMIL CU GND
@ 90°C RATED
IN 4" CONDUIT

MV

(3) 4/0 AWG AL (MV-105)
15 kV CLASS, TR-XLPE 100%
INSULATED 3Ø CONDUCTORS,
#1 AL EGC IN 5" CONDUIT

OHE

(3) 1/0 AAC (POPPY)
3Ø, 3W CONDUCTORS
WITH #2 AAC (IRIS) NEUTRAL

DC LINE DIAGRAM

1. Consistent Inverter Callout
2. Correct Inverter Topology from Manual
3. Inverter OCPD and Disconnect Type and Rating
4. String Fuse, DCB Fuse Size
5. Show the applicable schedules
    5.1. MPPT Schedule
    5.2. Harness Schedule

5.3. DCB/Inverter Schedule

6. Grounding Callout and Notes:
   6.1. PV Module Grounding Note
   6.2. Rack to CAB
   6.3. CAB to DCB/Inverter

7. Aux Cabinet for Central Inverter

8. Transformer Callout and Configuration for Central Inverter

9. Notes:

   9.1. PV Wire stranding note
   9.2. Module connectors note

RELAY and Inverter SETTINGS

1. Does the utility have specific relay settings? Is the utility NGrid, they have their own table that should be used.
2. Does the CT ratio match the recloser submittal, if no submittal default to 300:1.
3. VT Ratio should be defaulted to 234.5 for Tavrida recloser (Most commonly purchased by our customers).
4. Line to Line Voltage should match SLD POI voltage.
5. AC system size should match.
6. FLA should match SLD after XFMR.
7. Has electrical studies been conducted?
   a. If yes, then you should see NON-IEEE Functions populated by the electrical study's coordination curves. These curves should be pasted into the spreadsheet.
   b. If not, this section should be hidden.

CUSTOMER RECLOSER SETTINGS

Line-Line Voltage:	34500	
Line-Ground Voltage:	19919	
VT Ratio (X : 1) (effective ratio if LEA):	234.5	
VT Sec. V:	84.94	
CT Ratio (X:Y):	100	1
AC MW:	3.00	
Operating Power Factor:	1.00 absorbing / consuming	
AC MVA:	3.00	
FLA:	51	

Shall Trip Function	Device #	Recloser - SEL-651R			
		Voltage (L-N)			Clearing Time (s)
		(pu. of nominal voltage)	V (sec)	V (pri)	(s)
OV2	59-2	1.20	101.93	23903	0.32
OV1	59-1	1.10	93.44	21911	4.0
UV1	27-1	0.88	74.75	17529	4.0
UV2	27-2	0.50	42.47	9960	0.32
		Frequency (Hz)			Clearing Time (s)
OF2	81O-2	62.0			0.32
OF1	81O-1	61.2			600.0
UF2	81U-2	58.5			600.0
UF1	81U-1	56.5			0.32

Return to Service				
Min V	(27)	0.95	81	300 s
Max V	(59)	1.05	89	300 s
Min F	(81U)	59.5		300 s
Max F	(81O)	60.5		300 s

Non - IEEE 1547 Functions

Function	Device #	Setting	Amps - Primary	Amps - Secondary	Clearing Time (s)
Inst OC	50P	7.0	700	7	0.01
Time OC	51P	1.25 X FLA, U4 Curve, Time Dial = 2.2	64	0.64	2.0
Time Gnd OC	51G (or 51N)	U4 curve, Time Dial = 1.81	20	0.20	2.0

8. Check that the correct inverter table has been used. There is a specific table for NGrid.
    a. The Category I, II, and III options from IEEE 1547 are shown for completeness. The IEEE tables in this column are replicated for completeness because some values are fixed and some are a range and in the set of tables on the left we have to pick a setting from the range.
    b. If no direction is given in the CESIR, just use Category I
9. AC output L-L should match the inverter voltage rating.

INVERTER SETTINGS

Inverter AC Output L-L Volt. (V):	600
Inverter AC Output L-N Volt. (V):	346
Operating Power Factor:	1.0 absorbing/consuming

Shall Trip Function	Device #	Inverter Settings (same as IEEE 1547 Category I default Tables 11 & 18)		
		Voltage (L-N)		Clearing Time
		(pu. of nominal voltage)	(V)	(s)
OV2	59-2	1.20	720	0.16
OV1	59-1	1.10	660	2.0
UV1	27-1	0.70	420	2.0
UV2	27-2	0.45	270	0.16
		Frequency (Hz)		Clearing Time (s)
OF2	81O-2	62.0		0.16
OF1	81O-1	61.2		300.0
UF2	81U-2	58.5		300.0
UF1	81U-1	56.5		0.16

THREE LINE DIAGRAM

1. CONSISTENT NAMES AND RATINGS ACROSS SITE PLAN, POLE LINE UP, SLD, 3LD AND POLE ELEVATION.

2. ALL SLD ITEMS APPLY HERE

3. Don’t include cable size, FLA etc.

4. Only show equipment label with rating

5. Show CTs, VTs arrangement for metering

6. Phase of the Breaker is correct 3P vs 2P vs 1P

7. Transformer

    7.1. LV and HV Winding
    7.2. Transformer Internal Fuse and Switch
    7.3. Transformer grounding Connections

AUX SLD

1. Consistent with DAS vendor Drawings
2. If single phase transformer NO WYE/DELTA
3. Aux Power 3P vs 1P
4. Check all aux loads
    4.1. List all Loads: DAS, Data controller, Tracker Controller, Outlet, UPS
    4.2. Voltage and Phase: 3P vs 2P-240 vs 1P-120
    4.3. Motor Power Requirements
        - 1 Motor per Tracker– Self Powered
        - 1 Motor per Tracker – Powered from Aux Panel
        - 1 Motor per array
5. Motor Current Requirements
6. How is the motor daisy chained?
7. Match Breaker and circuit grouping with Feeder Plan
8. Check Ratings and Equipment Topology
    8.1. Aux Panel
    8.2. Aux transformer
    8.3. Breakers
    8.4. Cable and Conduit Size
    8.5. For CAB - Insulation of motor wire sunlight resistant?
9. Cable Size consistent with Electrical Sheet?

COMMUNICATION DIAGRAM

1. Has an Also Energy report been submitted?
    a. If no, then use default drawing
        i. Are daisy chains of inverters consistent with feeder plan?
        ii. Does the amount of equipment match the number of pads?

    iii. All weather sensors accounted for?
    iv. There should be a UPS for DAS
    v. DAS should connect to POI through fiber and POI should connect to recloser through ethernet
    vi. Are the line types correct
  b. If yes, has the submittal been approved
    i. Report should replace default drawing

GROUNDING DIAGRAM

1. Note for E-300 sheet
2. Follow the GND circuit from rack all the way to POI
3. Does DC Part reflect the equipment configuration?
4. Relevant Notes and Code callouts
5. XFMR Primary and secondary connections
6. Check Bonding Jumper
7. Check GEC
8. Grounding Ring consistent with PAD Layout

INVERTER ZONE MAP

1. Scale and North Arrow
2. Modules and Equipment are Properly Faded
3. Callouts:
   3.1. Fence
   3.2. Access Road
   3.3. Equipment Pad
   3.4. Inverter Zone Labels

AC and DC FEEDER PLAN

1. Scale and North Arrow
2. Fade the PV racks
3. Check for visibility and clarity of the PLOT
4. Legend Matches Cables Used
5. Callouts:
   a. Fence
   b. Access Road
   c. Equipment Pad
   d. DCB and Inverter number
6. Show Trench Reference
7. Show CAB Fill Reference

8. Identify Worst case for CAB fill
9. Show arrows for circuit count
10. Check Sheet Notes for circuit arrows
11. Follow each circuit and make sure it is correct
12. Show Cables for tracker Motor

COMMUNICATION FEEDER PLAN

1. Freeze the string wires and callouts
2. Fade the PV racks
3. Scale and North Arrow
4. **LIST THE INDIVIDUAL SENSOR NAMES AND LOCATION FROM DAS**
5. Check for visibility and clarity of the PLOT
6. Legend Matches Cables Used

EQUIPMENT AREA FEEDER PLAN

1. Fade the equipment
2. Show AC, DC, MV and Comms wiring
3. Check AUX rack with DAS vendor drawings
4. Aux Power
5. Clearance: NEC and Equipment
6. Inverter/Aux Rack Pile:
    - 6.1. Pile Count
    - 6.2. Pile Spacing
    - 6.3. Pile Type
    - 6.4. Pile Depth
7. Is Pad individual or combined? Do we have gravel?
8. Min clearance from side, front and back for inverter
9. Check NEC Table for clearance

TRENCHING DETAILS

1. Trench Depth vs Circuit Voltage
2. Conduit Spacing vs Count – Match with Feeder Plan
3. Direct Buried vs Conduit
4. Dimensions clearly shown

5. Tally with Ampacity Study
6. Check Notes
7. Check if you are capturing worst case – Don't pick easy drafting
8. Sheet reference to Feeder Plan
9. Callouts in correct places
10. Check Trench Headers and Match with Feeder plan
11. Check comms cable
12. Show GND wire in the conduct

ELECTRICAL SHEET

CHECK ALL THE HEADERS SO THAT THEY MAKE SENSE

1. PV System Parameter
    - 1.1. Weather Data Source
    - 1.2. Check PV Module Values
    - 1.3. Check Inverter Values
    - 1.4. Do you show all cases for mod per string and strings per inverter?
    - 1.5. Check SAM for bifacial module
    - 1.6. Is SAM value under inverter's limit?
    - 1.7. Is stringing, Ok? V, I, P, MPPT, No of Inputs, Temperature
    - 1.8. Does Total System Value match the SUMMARY TABLE?

2. Strings Per Raceway
    - 2.1. Use SAM Isc if available
    - 2.2. Add note for SAM sheet
    - 2.3. Only use single 1.25 for derate factors
    - 2.4. Confirm Temp and Fill derate factors
    - 2.5. Add Table for Free Air and Conduit if applicable
    - 2.6. Max Strings per conduit – Show all cases

SAM Isc (A)	NEC Isc Adj. Factor	Adj. Isc (A)	# Strings	# Conductors	Conductor Size	CU or AL	Voltage Rating (V)	Insulation Type	Temp. Rating (°C)	Circuit Limiting Raceway	NEC Ampacity (A)	Ambient Temp. (°C)	Derate - Temp.	Derate - # Conductors	Derate - Total	Final NEC Ampacity (A)	Ampacity Margin	Conduit Size	Conduit Fill
15.636	1.25	19.5	2	4	#10	CU	2000	PV	90	Conduit	40	33	0.96	0.8	0.768	20.4	49%	1"	31.58%
15.636	1.25	19.5	6	12	#10	CU	2000	PV	90	Conduit	40	33	0.96	0.5	0.48	32.6	19%	1-1/2"	38.10%
15.636	1.25	19.5	10	20	#10	CU	2000	PV	90	Conduit	40	33	0.96	0.5	0.48	32.6	19%	2"	37.80%
15.636	1.25	19.5	15	30	#10	CU	2000	PV	90	Conduit	40	33	0.96	0.45	0.432	36.2	10%	2-1/2"	39.57%

    - 2.7. For mix circuits Conduit and Free Air – Check 10 ft rule

3. String, Harness and DCB Schedule
    - 3.1. Spot check distances for excel error
    - 3.2. Show SAM Isc if available
    - 3.3. Edit header for STRING INVERTER vs DCB
    - 3.4. Add a row for Total Strings – Match with summary table.
    - 3.5. Check Termination Rating

- 3.6. Equivalent OCPD for DCB
- 3.7. Tally I/O Cable Limits with inverter and DCB
- 3.8. Cable Sizing
    - 3.8.1. No of Sets
    - 3.8.2. 2 wires for DC
    - 3.8.3. Cable size
    - 3.8.4. AL vs CU
    - 3.8.5. Cable Insulation
    - 3.8.6. GND Wire Size – No upsize

- 3.9. Conduit Sizing
    - 3.9.1. Conduit Material – Mix?
    - 3.9.2. Direct Buried
    - 3.9.3. Conduit Fill

4. AC Circuit Schedule
    - 4.1. Spot check distances for excel error
    - 4.2. Check Headers
    - 4.3. Check Termination Rating
    - 4.4. Check Voltage, FLA
    - 4.5. OCPD Below vs Above 800A
    - 4.6. Cable Sizing
        - 4.6.1. No of sets – Don't go equipment Limit
        - 4.6.2. No of wires per set – 4 vs 3 vs 2 – Check phase of circuit
        - 4.6.3. Wire Size – Use 75C
        - 4.6.4. Cable Insulation > Circuit Voltage
        - 4.6.5. Terminal Rating
        - 4.6.6. GND Wire Size – Upsize for Phase Conductors
        - 4.6.7. Cable Derate
    - 4.7. Conduit Sizing
        - 4.7.1. Conduit Material – Mix?
        - 4.7.2. Direct Buried?
        - 4.7.3. Conduit Fill < 40%

5. Aux Circuit Schedule
    - 5.1. Follow Point no 4
    - **5.2. No of wires per set – 3 vs 2 – Check phase of circuit and Voltage**
    - 5.3. Tally Aux loads with SLD
    - 5.4. Tally Aux Loads with DAS vendor Drawings
    - 5.5. Check Motor Power requirements
    - 5.6. Voltage Drop of Motor < Allowable Circuit Voltage

6. MV Circuit Schedule

6.1. Check Termination Rating
6.2. Check Circuit Voltage
6.3. 1.25xFLA < Wire Rating – Use Joe's handbook
6.4. Tally Wire Size with SLD
6.5. Check GND Wire – No Concentric Neutral for EGC
6.6. Conduit Sizing
6.6.1. Conduit Material – Mix?
6.6.2. Direct Buried?
6.6.3. Conduit Fill < 40%

7. Use appropriate VD summary- Don't leave everything blank

Average Voltage Drop Summary	DC	LV-AC	MV
Calculated VD	1.16%	0.26%	0.09%
TOTAL AC-DC VD	1.51%		
Client Criteria	3%		

Voltage Drop Summary	DC	AC	AC and DC
Calculated VD	0.55%	0.88%	0.71%
Calculated worst case VD	0.57%	1.32%	0.95%
Client Max Average VD	N/A	N/A	3%

ELEVATION DETAILS

1. Confirm if structural is complete?
2. Check Pile Spacing, Type, Count and Depth
3. Clearance from Grade
4. Check callouts
5. Sweep/ Bends Distance
6. Match with SLD
7. Min clearance from side, front and back for inverter
8. Check NEC Table for clearance
9. Match weather sensors with DAS and Comms SLD
10. CAB Spacing, SAG, Clearance from Grade

CAB or Cable Hanger DETAILS

1. Calculate CAB section weight
2. Check SAG
3. Check CAB section Fill
4. Check conductor grouping – Voltage vs DC/AC vs Comms
5. Check Standard Details
6. Check Pile depth
7. Does CAB arrangement make sense?

PAD – SLAB DETAILS

1. Show rebar schedule

OVERALL SITE GROUNDING PLAN

1. Fade the PV racks
2. Check consistency with grounding SLD
3. Use latest template

POLE DETAILS

1. Use latest template
2. Consistent Names
3. Match Configuration
4. Remove unnecessary details

LABELS

1. Check NEC reference
2. Voltage and Current match calc sheet
3. GOAB = Main Service Disconnect
4. Check Client’s Name and Phone Number

PVSyst Analysis Summary

1. Project Info

- Location
- System Type: Fixed Tilt or Tracker
- Size: DC & AC capacities

2. Equipment

- Module: Type and correct count
- Inverter: Type and correct count

3. Losses

- Soiling Loss: 3% (default)
- Thermal Loss: Uc = 25, Uv = 1.2
- Wiring Loss: E-300 in Planset (default total ~1.5%)
- Albedo: Use SolarAnywhere or get from SAM
- Rack Height: Check Elevation Detail (default 1.8m if not provided)
- Other Losses: LID, quality, mismatch — use PVSyst defaults if no data

4. Shading & Simulation

- Near Shading: “Fast” method, 80% electrical effect (except First Solar)

5. Advanced

- Backtracking: Enable if using trackers
- Bifacial: Enable if using bifacial modules